Information Management for CS
SE 2: Information Systems and the Internet

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Slides based on Prof.dr.ir. Hajo Reijers

Plan for lectures

1. Information and Strategic planning
2. Information Systems and the Internet
3. Guest Lecture: Digital Process Excellence (KPMG)
4. Information Systems Development
5. Business Processes and Information Systems
6. Petri Nets
7. Analysis with Petri Nets
8. Real life/complex Petri Nets

Agenda

1. Introduction to e-Business and e-Commerce
2. The Cloud
3. Business Intelligence

Lecture 1: Questions

Refresher question

1. Business processes can be broadly categorized into strategic, tactical, and operational.
   i) Briefly describe each category
   ii) For each category, provide an example process from the Bike Rental information system discussed in the course.

Exemplary Exam Question

In which stage of the IS strategy development is the vision of the future direction of the organization discussed?

a) Strategic Objective
b) Strategic Analysis
c) Strategic Implementation
d) Strategic Definition

Introduction to e-Business and e-Commerce

E-Business

- Using web technology to
  - Improve business processes
  - Enhance communication with partners, suppliers, vendors and customers
  - Perform business transactions securely

e-business (e' biz' nis): The process of using Web technology to help businesses streamline processes, improve productivity and increase efficiencies. Enables companies to easily communicate with partners, vendors and customers, connect back-end data systems and transact commerce in a secure manner.

- Potential areas in the business:
  - Production processes: *procurement, ordering stock, payment, supplier communication, production control*
  - Customer-focused processes: *marketing, selling, customer support, order and payment processing*
  - Internal management processes: *training, recruitment, employee services*

Definition: E-Commerce

- E-Commerce can be defined as:

  "any exchange of information or business transaction that is facilitated by the use of information and communication technologies"

- E-Commerce goes beyond home computer users
- Most parties involved are companies and public authorities
- The variety of systems is huge

Types of e-commerce activities

Transaction Type	Example
B2B	Transactions between businesses
B2C	Companies selling products to customers
B2G	Transactions between businesses and public sector organizations
C2C	Transactions between private individuals

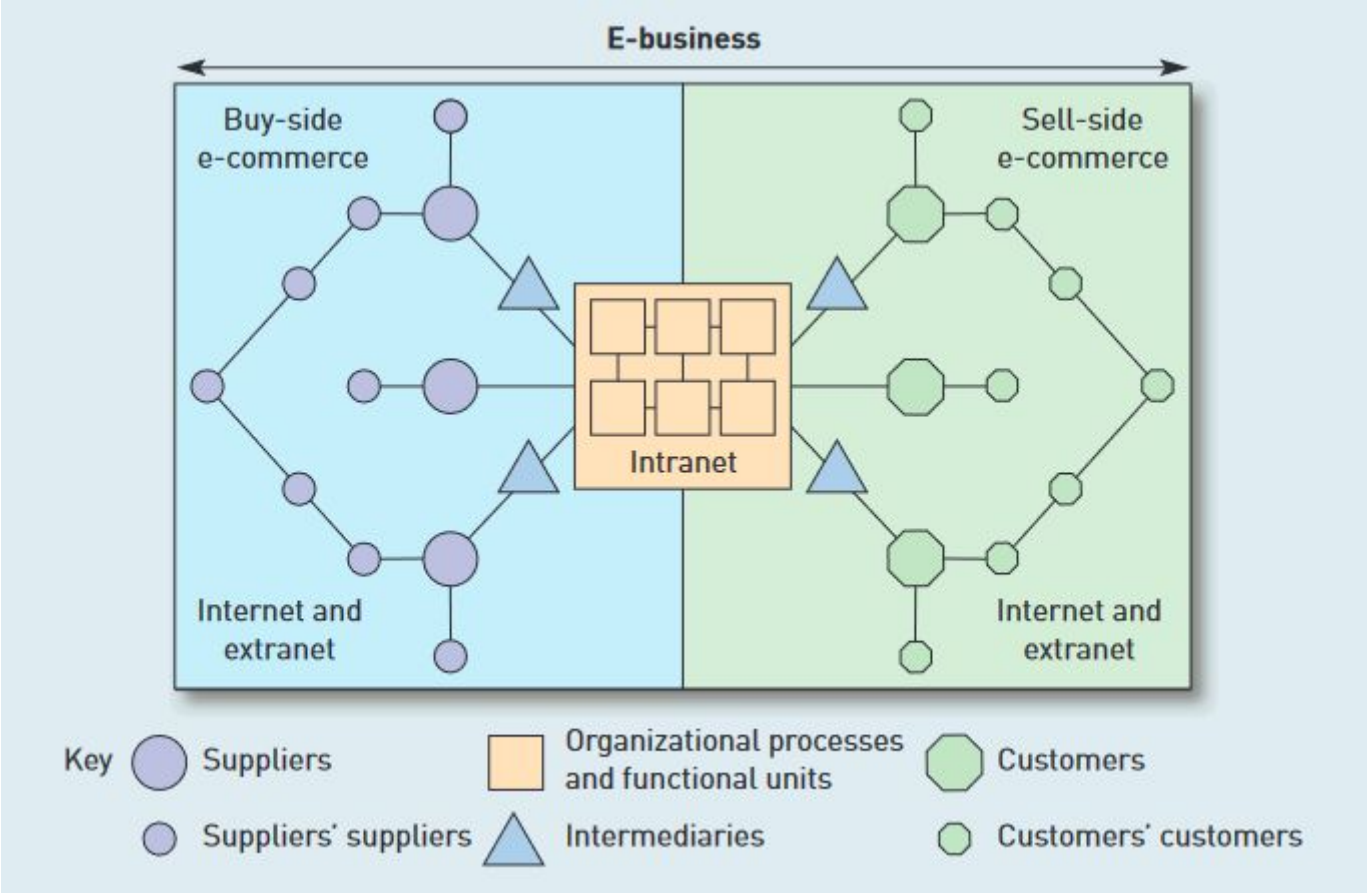
E-Commerce Systems

- E-shops and e-malls
- E-procurement
- E-auctions
- Content providers
- Market segmenters, infrastructure providers
- Specialist service providers

Drivers for Using the Internet for Business

- Cost
- Flexibility
- Protecting investment
- Connectivity
- Low risk
- Improved customer service
- Globalization

Relationship between E-commerce and E-business



Benefits of E-business and E-Commerce

Tangible benefits	Intangible benefits
■ Increased sales from new sales leads, giving rise to increased revenue from: – new customers, new markets – existing customers (repeat-selling) – existing customers (cross-selling) ■ Marketing cost reductions from: – reduced time in customer service – online sales – reduced printing and distribution costs of marketing communications ■ Supply-chain cost reductions from: – reduced levels of inventory – increased competition from suppliers – shorter cycle time in ordering ■ Administrative cost reductions from more efficient routine business processes such as recruitment, invoice payment and holiday authorization	■ Corporate image communication ■ Enhancing brand ■ More rapid, more responsive marketing communications including PR ■ Faster product development lifecycle enabling faster response to market needs ■ Improved customer service ■ Learning for the future ■ Meeting customer expectations to have a website ■ Identifying new partners, supporting existing partners better ■ Better management of marketing information and customer information ■ Feedback from customers on products

The Cloud

There is no cloud
it's just someone else's computer

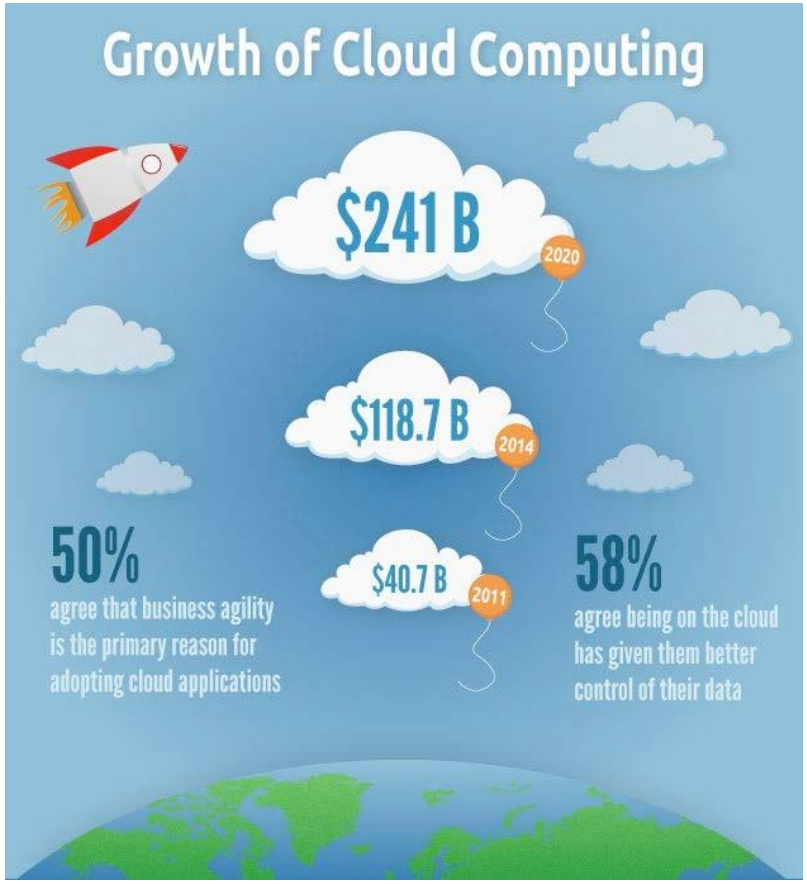
What is Cloud Computing?

- Services and Solutions that are delivered or consumed in real time over the Internet

- Cloud computing is a delivery model of computing services over the Internet
- It enables real-time development, deployment, and delivery of a broad range of products and services

Why is the Cloud important?

- We face an increasing importance of cloud computing
- Gartner sees the cloud as one of the top 5 technology trends in the next years
- How does cloud computing change business?



Impact of Cloud for Business

The Established Ones

IBM

Google

SAP

ORACLE

Microsoft

amazon

The Challengers

salesforce

workday

box

servicenow

NETSUITE

xero

What is SaaS?

SaaS – software-as-a-service – is a way of delivering applications over the Internet. Customers can access SaaS applications right from a Web browser, which means there is no hardware or software to buy, install, maintain, or update. The SaaS provider takes care of everything – and the customer always has the latest version of the application.

Explore our SaaS products:

- SAP SuccessFactors HR Solutions
- SAP S/4HANA Finance
- SAP S/4HANA Cloud (Cloud ERP for large enterprise and midsize business)
- SAP Hybris Cloud for Customer
- SAP Hybris Sales Cloud
- SAP Hybris Service Cloud
- SAP Hybris Marketing
- SAP Hybris Commerce
- SAP Business ByDesign (Cloud ERP for Midsize Business)
- SAP Business One Cloud (ERP for Small Business)
- SAP Analytics Cloud
- SAP Jam Collaboration
- SAP Jam Communities
- SAP Integrated Business Planning
- Ariba Network: Sourcing, Procurement, & Finance
- Concur: Travel & Expense Management
- Fieldglass: Flexible Labor Management

Characteristics of the Cloud

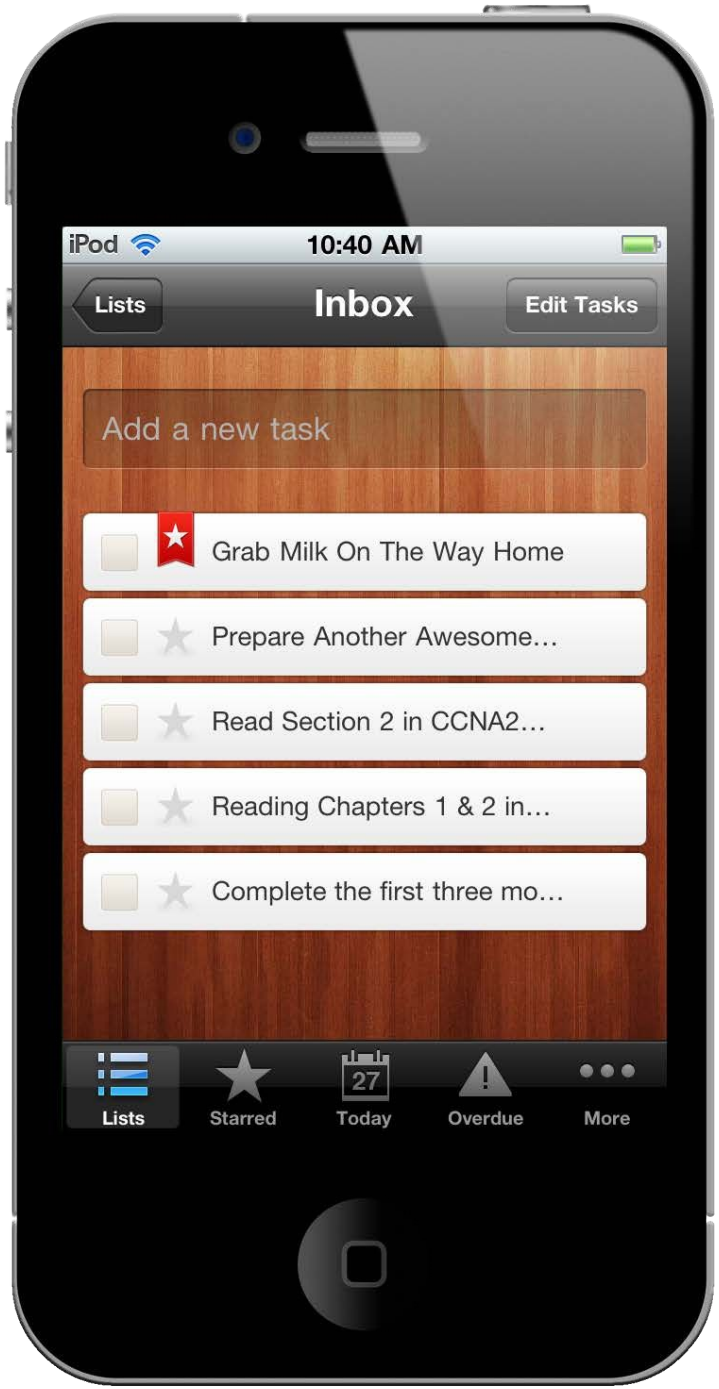
- Providing services via the cloud is different from on-premise solutions
- Important characteristics of cloud solutions:

1. Ease of use
2. Specific for one purpose
3. Simplicity
4. Collaboration
5. Mobile Access
6. Customer Success Management

1. Ease of use

Dropbox

Simplify your life.


iPod
10:40 AM
Lists
Inbox
Edit Tasks
Add a new task
Grab Milk On The Way Home
Prepare Another Awesome…
Read Section 2 in CCNA2…
Reading Chapters 1 & 2 in…
Complete the first three mo…
Lists
Starred
Today
27
Overdue
More

2. Specific for one purpose

timeoffmanager

Home | Users | Settings | Reports | Reminders | Tools | Account | Help | Logout

Jason Gendron
President
Ironflow Technologies Inc.
jasong (administrator)
Account Status : Active
Next Billing Date: Dec 31, 2011

1

Quick Stats

Pending Requests	33
Employees Off Today	0
Employees Off This Week	7
Employees Off Next Week	0
Total Users	24

2

Pending Requests (33) | Banked Time (2) | Cancellations (3) | Notifications (0)

Employee	Type	StartDate	Length	Days	Details
John Smith	Appointment	01/12/2011	4.000h	Partial	
Hieg Khatcherian	Appointment	01/13/2011	2.000h	Partial	
John Smith	Appointment	01/13/2011	4.000h	Partial	

3

November 2011 | today | day | week | month | Expand Cells

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
30	31	1	2	3	4	5
		John Smith			Justin Martin	
6	7	8	9	10	11	12
	Justin Martin	Jacob Lagacy		Mark Murray		
13	14	15	16	17	18	19
		Justin Martin		Robert Allan Rachel Cyr		
20	21	22	23	24	25	26
	Justin Martin Jacob Lagacy		Michelle Smith	John Smith Justin Martin Mark Murray MORE		
27	28	29	30	1	2	3

workable DASHBOARD CANDIDATES AGENDA REPORTS ACTIVITY Search...

Head of Account Management

+ Add candidates | Configure

SOURCED | 13 APPLIED | 3 PHONE SCREEN | 2 INTERVIEW | OFFER | HIRED

QUALIFIED | DISQUALIFIED 11

Select all | Bulk actions

Filter by name, headline or #tag

Mike Bowder
Account Director at JBox Media
#accountant #director #experienced
from twitter.com · 2 days ago

Sylvia Hardin
Experienced Sales Manager
#sales #manager #experienced
from indeed.com · 4 days ago

Laura Schmidt
Country Manager (UK) at Sponge
#manager #sales #experienced
from indeed.com · 5 days ago

John Terry
Experienced Sales Manager
#commercial #sales #manager
from indeed.com · 5 days ago

Cindy Sawyers
Country Manager (UK) at Sponge
#manager #sales #commercial
from twitter.com · 5 days ago

George Linnert
Commercial Director at Cellcom
#commercial #director #experienced
from linkedin.com · 5 days ago

PHONE SCREEN

Sylvia Hardin
Experienced Sales Manager
#sales × #manager × #experienced × + add tag

Candidate Profile

WORK EXPERIENCE (about 8 years total)

2012 - now	**VP Sales, Europe** at **GeoTap Software** (about 2 years)
2010 - 2012	**Sales Director** at **Hoyle Systems** (about 2 years)
2008 - 2010	**Business Development Manager** at **Hoyle Systems** (about 2 years)
2006 - 2008	**Account Manager** at **Logicon Ltd** (about 2 years)

EDUCATION

2011 - 2014	B.A. Marketing Communications at **University of Manchester**
2008 - 2011	MBA at **University of Manchester**

CONTACT DETAILS

Address	642 Ernser Parkway, UD2B 5AB, Lonn, UK
Phone	0161 195 4753
Email	ryann@heathcote.info

Answers

peopleperhour

POST PROJECT | SEARCH | HOW IT WORKS | LOG IN | SIGN UP | FREELANCER?

Curated Freelancers — 31 found
Hire Freelance English to Greek Translators

Search for any skill

In: English To Greek

Filters | Industry | Location | Language | Popularity | CERT | Rate | Available Now | Has Portfolio

Fani K. CERT 5

EN / DE / EL Translator - Interpreter

ATHENS, GR

Greek <=> English | editing | project management | + 10 MORE SKILLS

5.0 (27) €15/hr

CONTACT

VIEW PROFILE

Michalis X. CERT 4

Translator Transciber

Michalis X.

IRAKLEION, GR

Greek <=> English | proofreading | German <=> English translation | + 2 MORE SKILLS

5.0 (29) €12/hr

CONTACT

VIEW PROFILE

Vassilis N. CERT 4

Translator, Interpreter, Editor

ATHENS, GR

editing | proofreading | tutorial | + 1 MORE SKILL

4.9 (15) €29/hr

CONTACT

VIEW PROFILE

3. Simplicity

One Size Fits Most

When you buy a SaaS solution, you should care more about the neighborhood than the house

4. Collaboration

Overleaf

ICSE_2019_SIP

Source | Rich Text

Recompile

```
14  To implement this
    science-centric vision,
    Netflix's experimentation
    platform, \texttt{Netflix
    XP}, was reimagined around
    three key tenets:
    trustworthiness,
    scalability, and
    inclusivity. Trustworthiness
    is essential since results
    that are untrustworthy are
    not actionable. Scalability
    is required to accommodate
    for Netflix's growth.
    Inclusivity is a key tenet
    because it allows scientists
    from diverse backgrounds
    such as biology, psychology,
    economics, mathematics,
    physics, computer science
    and other disciplines to
    contribute to and enrich the
    experimentation platform.
15
16  The implications of these
    tenets on \texttt{Netflix
    XP} are wide-ranging, but
    perhaps chief among them are
    the resulting choices of
    language and computing
    paradigm. Python was chosen
    as the primary language of
    the platform; with some
```

Track changes is off

science-centric from our perspective is about the emphasis on science/research in the platform, rather than on the *scientist* in the platform. I think as soon as the vision focuses on workflows in Netflix XP, it actually goes against what David was saying. If we removed the word "platform" from my original suggestion, does that address David's point?

Oct 4, 2019 8:57 PM

You: OK sounds good. The original idea was to somehow get across the idea/motto "we optimize for people than for performance". But I agree this message was convoluted with all the changes. I like your suggestion without the platform!

Oct 5, 2019 8:38 AM • Edit • Delete

Hit Enter to reply

Resolve | Reply

Engineering for a Science-Centric Experimentation Platform ICSE '20, May 23-29, 2020, Seoul, South Korea

3 RESEARCH METHOD

4 SOFTWARE ARCHITECTURE

QuickBooks.

Joe's Landscaping

- Home
- Customers
- Vendors
- Employees
- Transactions
- Reports
- Taxes
- Apps

Customers

New customer

Unbilled		Unpaid		Paid
$325 2 ESTIMATES	$375 10 UNBILLED TIME & COS	$1,390 10 OPEN INVOICES	$1,312 5 OVERDUE	$570 14 PAID LAST 30 DAYS

Batch actions | Sort by name | Find a customer

CUSTOMER	PENDING INVOICES	BALANCE	ACTION
A Rental	1 Open invoice	$54.00	Receive payment
Amy's Bird Sanctuary Amy's Bird Sanctuary (650) 555-3311	1 Open invoice	$25.00	Receive payment
Bill's Windsurf Shop Bill's Windsurf Shop (415) 444-6538			Create invoice
bridgette haymaker	1 Open invoice	$75.60	Receive payment
Cool Cars Cool Cars			Create invoice

5. Mobile access

Remember everything | Evernote Corporation

http://www.evernote.com/about/home.php

Apple Yahoo!

EVERNOTE

PRODUCTS | NOTEWORTHY BLOG | THE TRUNK

- WEB SIGN IN
- CREATE ACCOUNT
- GO PREMIUM

Remember everything.

Capture anything.

Save your ideas, things you like, things you hear, and things you see.

Access anywhere.

Evernote works with nearly every computer, phone and mobile device out there.

Find things fast.

Search by keyword, tag or even printed and handwritten text inside images.

GET EVERNOTE, IT'S FREE ▸

Also by Evernote:

Skitch ▸

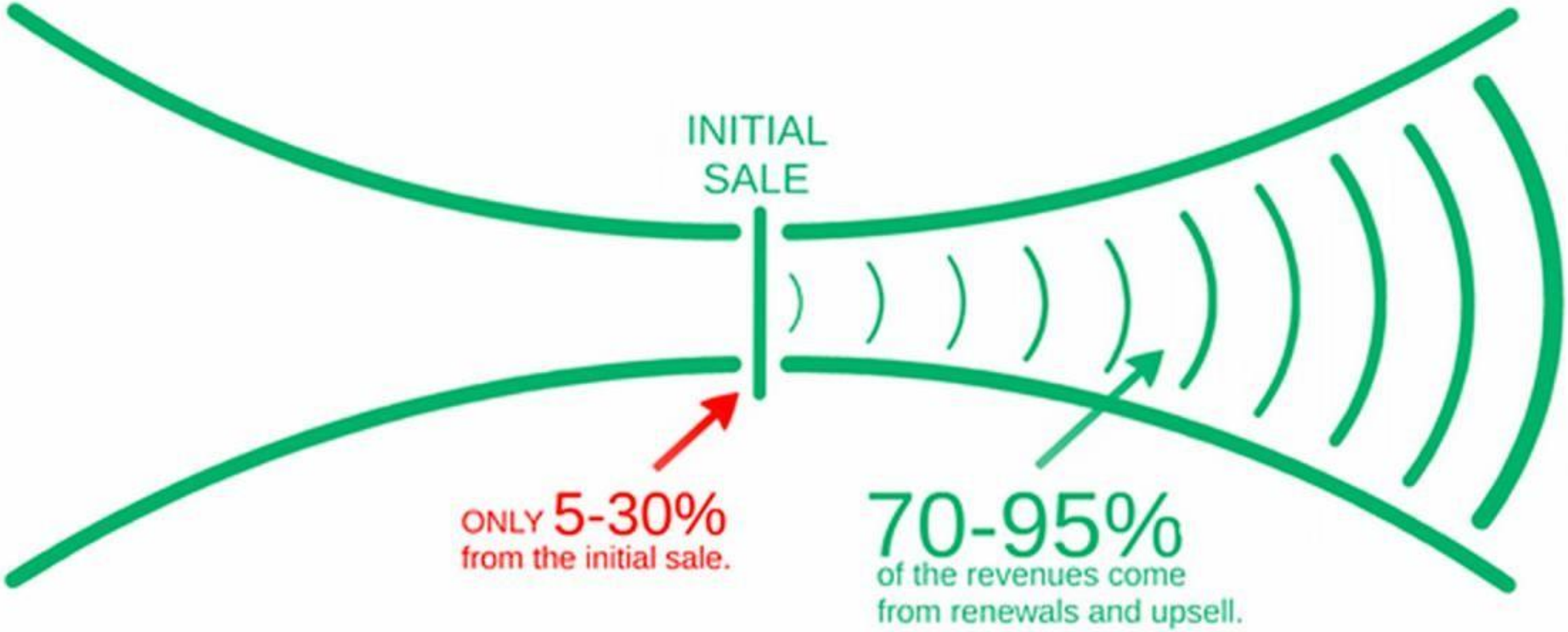
Evernote Hello ▸

Evernote Food ▸

Evernote Clearly ▸

Evernote Peek ▸

6. Customer success management


INITIAL
SALE
ONLY 5-30%
from the initial sale.
70-95%
of the revenues come
from renewals and upsell.

FINDING NEW CUSTOMERS PROCEDURE
KEEPING OLD CUSTOMERS PROCEDURE
GOFF
AllPosters

How does the cloud change business?

- Let us consider an example from the business process modeling domain

ARIS Platform

vs.

SIGNAVIO
collaborative process design

ARIS

- Over 20 years old
- 200,000 ARIS community members
- 2,000,000 ARIS users
- Leader in all respective market areas
- Highest market share of any modeling tool

Not the kind of company you want to compete with!?

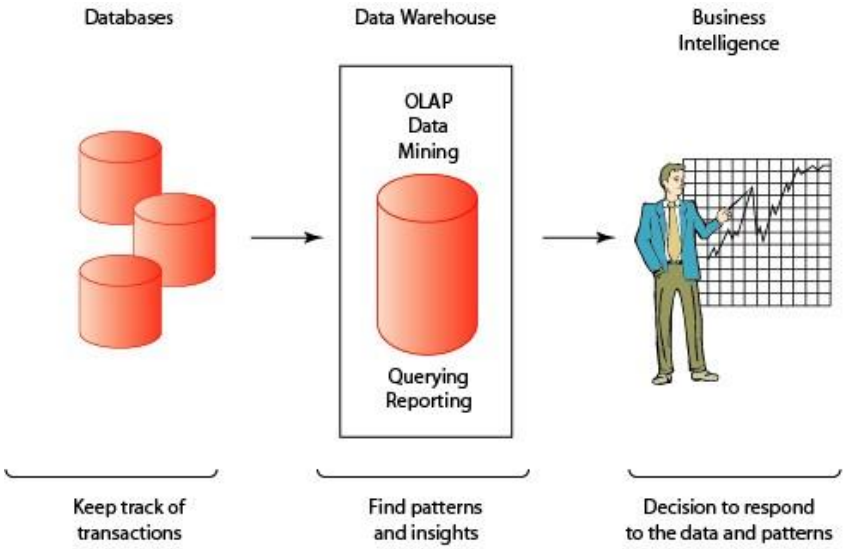
Signavio

- The world‘s first cloud-based process modeling tool
- Founded 2009 (1st prototype 2006)
- In 2011: "ICT startup of the Year” in Germany
- In 2018: Inc. 5000 Europe Ranking
- 75% annual growth, 250 employees
- More than 1000 customers:

Business Intelligence

Business Intelligence for Decision Making

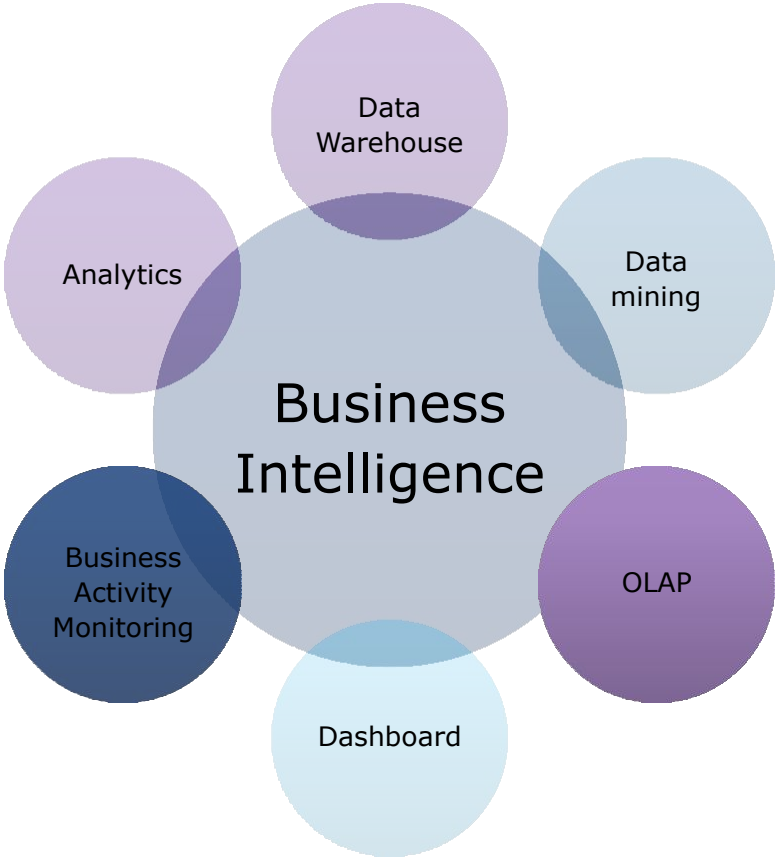
- Business intelligence:
  - *"the result of applying **tools and techniques** to the **data and information** available within and without an organization to facilitate **a better understanding** of both its environment and its operation and thereby improve the **decision-making process**."*



Key Features of Business Intelligence

- Internal and external inputs
- Structured and unstructured data
- Improved support for creating business strategy
- Gaining competitive advantage
- Improving support for decision making
- Increasing the use of performance indicators
- External drivers

Elements of BI Systems



Business Intelligence in Action

https://www.youtube.com/watch?v=AJ46psRBTj4

Tools and Techniques for Business Intelligence

- Some common tools and techniques to generate business intelligence
  - *Decision support systems*
  - Group decision support
  - Document management systems
  - Online analytical processing (OLAP)
  - Data warehousing
  - Data mining
  - *Digital dashboards*

Expert systems

- A computer system that emulates the decision-making ability of a human expert
- A big hype in the 1980s, almost vanished during the 1990s
- Revival nowadays through Robotic Process Automation:
  - A software application that replicates the actions of a human being interacting with the user interface of a computer system

ELIZA Terminal

```
Welcome to
                  EEEEEE  LL      IIII  ZZZZZZZ   AAAAA
                  EE      LL       II       ZZ   AA   AA
                  EEEEE   LL       II     ZZZ    AAAAAAA
                  EE      LL       II    ZZ      AA   AA
                  EEEEEE  LLLLLL  IIII  ZZZZZZZ  AA   AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.


ELIZA: How do you do.  Please tell me your problem.
YOU:
```

This page is a sample for mass:werk termlib.js and the ELIZA library elizabot.js.

For a chat with two Eliza therapists see the ELIZA-Chat.

ELIZA is a natural language conversation program described by Joseph Weizenbaum in January 1966 [1]. It features the dialog between a human user and a computer program representing a mock Rogerian psychotherapist.

Compatibility:

This page uses JavaScript and DHTML and should be compatible with most standard web browsers.
Please note that khtml (Konqueror) currently does not handle any keyboard events.

References:

[1] Weizenbaum, Joseph "ELIZA – A Computer Program For the Study of Natural Language Communication Between Man and Machine"
in: Communications of the ACM; Volume 9 , Issue 1 (January 1966): p 36-45.

© Norbert Landsteiner 2005; <http://www.masswerk.at>

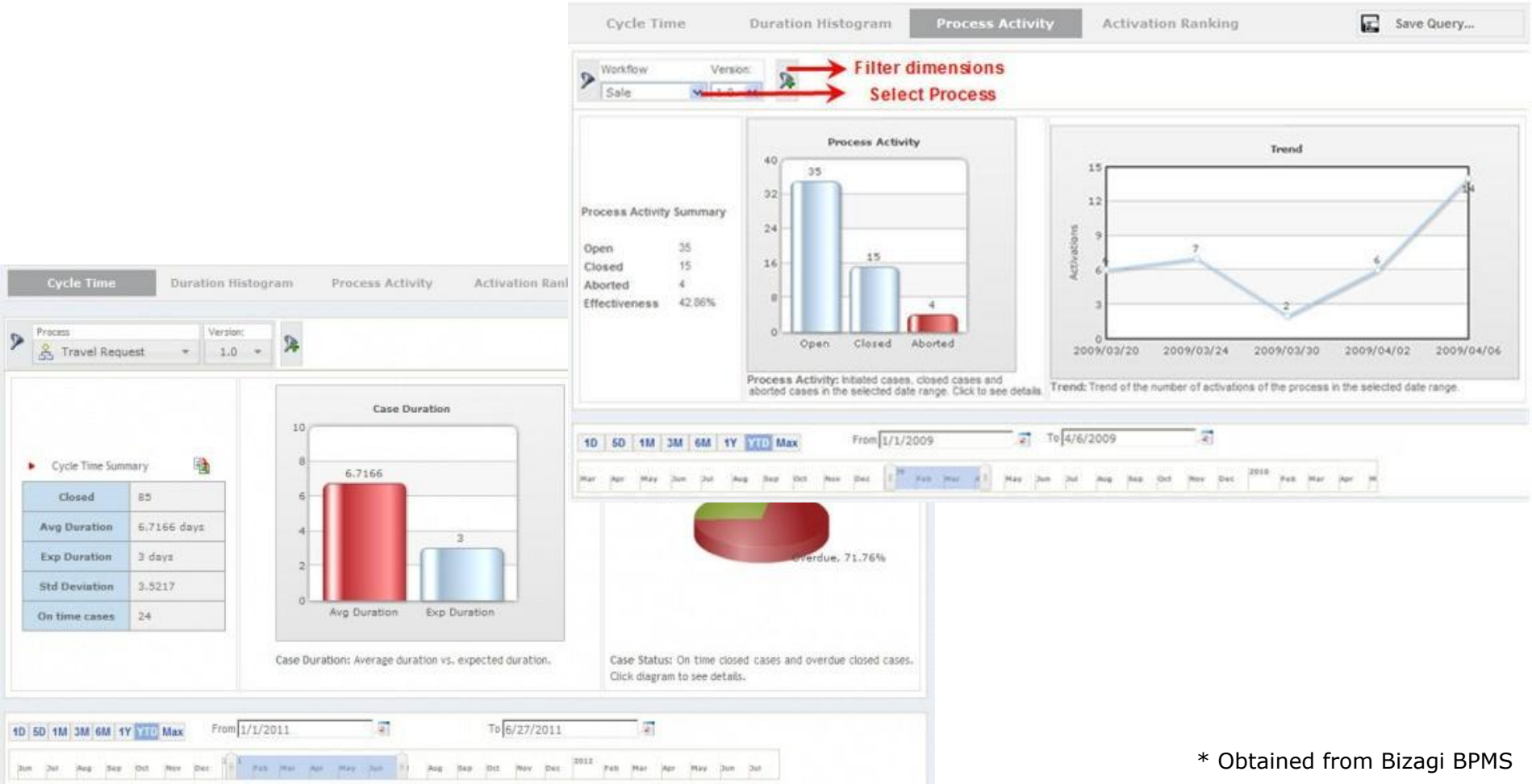
https://www.masswerk.at/elizabot/eliza.html

Process Intelligence in Business Process Management Systems

- Digital dashboards used for process monitoring and control
  - Runtime monitoring (Business Activity Monitoring)
  - Post-mortem monitoring (Business Process Analytics)

Business Process Analytics

- Dashboards for cycle time, duration histogram, process activity, activation ranking, frequent paths *



* Obtained from Bizagi BPMS

Summary

Summary

- Any business-related information exchange based on information and communication technology can be referred to as e-business
- E-business may change markets completely (Amazon, Ebay, Spotify, …)
- Business intelligence is key to better decision making based on internal and environmental data